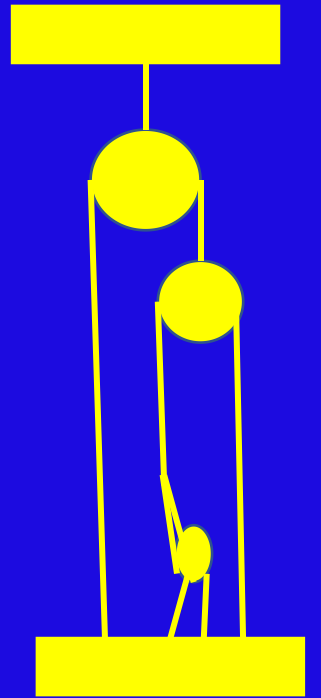


Quiz02: A person with $m_p = 60$ kg stands on a base plate with $m_b = 30$ kg. They are connected by a rope and pulley as shown in the figure below. Assuming that the mass of the pulley and the rope can be neglected and the friction at the pulley can be disregarded, and the rope is inextensible, if the person and the base plate are to rise with an acceleration of 1 m/s^2 , what is the force F_n exerted by the person on the rope? What is the pressure exerted by the person on the base plate? (Take $g = 10 \text{ m/s}^2$.)



Answer: $F = F'_T + 2F'_{T2} - (m_p + m_b)g = (m_p + m_b)a$

$$F'_T = 2F'_{T2}$$

$$F'_{T2} = 247.5N$$

$$F_N + F_{T2} - m_p g = m_p a$$

$$F_N = 412.5N$$

